Agentic DIY Project Companion

Product requirements, workflow, market analysis and delivery roadmap

Version 1.0 | Research date: 17 October 2026 | Initial geography: Germany

1. Executive decision

Build a project-completion assistant, not another generic DIY chatbot. The proposed app takes project photos, measured dimensions, user constraints and an inventory of owned items; turns them into a reviewed bill of materials; matches compatible products; identifies practical nearby shopping options; and supports purchase verification and assembly.

Recommended entry point: a German/English mobile-first prototype for bounded indoor shelf and furniture projects, with a safety-gated branch for wall mounting. Start with a curated catalogue and one cooperating store. Add broader retailer coverage only after product, stock and location data access is demonstrated.

The idea is not an uncontested market. heyOBI, HORNBACH and BAUHAUS already advertise in-store product-location capabilities. DIWAI and SnapToBuild advertise photo-to-project planning. Home Depot's August 2026 announcement describes photo uploads, project guidance, local availability, maps, aisle/bay locations and deployment in all 2,000+ US stores. Differentiation must therefore come from reliable compatibility validation, owned-item reconciliation, German localisation and the continuity of the complete project record—not the presence of agents alone. [web:6][web:36][web:38][web:64][web:65][web:81]

Evidence conventions

- Verified capability means described in an official or primary public source, not independently performance-tested.
- Advertised capability means a vendor claim that needs hands-on validation.
- Proposed requirements, costs, targets, timelines and commercial scenarios below are planning assumptions, not externally verified outcomes.
- Not verified means this research did not establish the capability; it does not prove that the capability is absent.
- No retailer API access, commercial agreement, inventory feed or map licence has been established by this research.
- “Tool” in the brief is interpreted provisionally as “toom”; confirm if a different business was intended.

2. Problem, users and value proposition

Jobs to be done

1. Translate “I want a shelf like this” into a feasible, bounded project.
2. Identify which dimensions, wall properties and load requirements still need confirmation.
3. Avoid buying incompatible screws, fixings, brackets, drill bits or tools.
4. Reuse suitable items already owned rather than replacing them unnecessarily.
5. Reduce time spent switching between advice, product pages, store searches and assembly instructions.
6. Arrive at a store with a usable list and truthful location guidance.
7. Finish safely, or recognise when professional help is necessary.

Initial personas — hypotheses to validate

- Beginner renter: limited tools, unfamiliar German hardware vocabulary, uncertainty about wall mounting and permissions.
- Occasional DIY homeowner: owns some equipment but needs compatibility and quantity checks.
- Multilingual shopper: can describe a project but struggles with local product terminology.
- Store associate: needs an explainable project summary to help a customer efficiently.

Value proposition

“From your project photos and what you already own to a checked shopping list and guided next steps.” Do not promise guaranteed structural safety, exact image-derived measurements, universal live inventory or navigation in every retailer.

3. Product scope

MVP includes

- Indoor freestanding shelf/furniture assembly and explicitly bounded shelf-mounting planning.
- Text, photos, barcode/model-number entry and manual measurements.
- German and English; metric units; EUR price presentation.
- Owned-item inventory with quantity, dimensions and confirmed suitability.
- Clarifying questions, generic bill of materials, catalogue matching and explainable alternatives.
- Store search by postcode or optional location, availability provenance, and retailer handoff.
- Aisle/zone guidance only where an authorised feed or verified pilot dataset exists.

- Project checklist, manufacturer-document links, offline access to saved lists and issue reporting.

Excluded from MVP

- Structural engineering design or load-bearing building alterations.
- Gas work, mains electrical installation, concealed plumbing repair, roof work and other high-risk trades.
- Automated drilling-site clearance from photographs.
- Automatic purchases, unapproved substitutions or autonomous payments.
- Guaranteed identification of screw grade, wall material, hidden wiring or water pipes from photos.
- General-purpose CAD, precise AR installation and indoor blue-dot positioning across all stores.

Wall mounting is a gated feature, not a low-risk default. If prerequisites cannot be established, the app may continue with general planning while withholding actionable fixing selection or drilling instructions.

4. End-to-end user workflow

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flowchart TD
 A[Create project: text, photos, constraints] --> B[Extract candidate objects and inv
 B --> C[User confirms measurements and owned items]
 C --> D{Safety-critical facts complete?}
 D -- No --> E[Ask targeted questions / request evidence / expert handoff]
 E --> C
 D -- Yes --> F[Generate generic BOM and dependency graph]
 F --> G[Validate materials, dimensions and compatibility]
 G --> H[Subtract confirmed suitable inventory]
 H --> I[Retrieve retailer product candidates]
 I --> J[Apply hard constraints; rank feasible baskets]
 J --> K[Check stores, stock evidence and data freshness]
 K --> L[User selects basket and fulfilment]
 L --> M[Retailer checkout or explicitly approved reservation]
 M --> N[In-store checklist and verified aisle guidance]
 N --> O[Scan purchased item; revalidate substitutions]
 O --> P[Manufacturer-grounded assembly with checkpoints]
 P --> Q[Completion, issues, remaining stock and feedback]
 G -- Conflict --> E
 O -- Mismatch --> I
```

Detailed steps and outputs

Step	User or system action	Required output / gate
1	Describe shelf, purpose, budget, tools and location preference	Project draft and scope classification

Step	User or system action	Required output / gate
2	Upload reference, room and owned-item photos separately	Candidate observations with uncertainty, not asserted facts
3	Enter width/depth/thickness, intended contents, mounting type	Confirmed specification; missing facts explicitly listed
4	Confirm wall substrate, bracket details and manufacturer information if mounting	Safety gate; professional referral if unresolved
5	Create materials, consumables, tools and PPE list	Dependency-linked generic BOM
6	Match owned items by quantity and specification	Suitable / uncertain / unsuitable classification
7	Search approved catalogues	Exact variant candidates with evidence
8	Validate combinations	Feasible basket or clear reason no match is possible
9	Compare shopping plans	Fewest trips / lowest cost / balanced options
10	Select store and fulfilment	Timestamped availability and explicit limits
11	Approve external action if offered	Exact retailer, variant, quantity, price and action
12	Navigate using verified locations	Aisle/zone list or honest fallback
13	Scan item and handle substitution	Compatibility checked again before acceptance
14	Follow assembly sequence	Manual-grounded steps with stop conditions
15	Record completion or unresolved issue	Project history and inventory update

5. Functional requirements and acceptance criteria

P0 = MVP mandatory; P1 = pilot extension; P2 = later.

ID	Priority	Requirement	Acceptance criterion
FR-01	P0	Create, save, resume and delete projects	A returning user can resume the last confirmed project state
FR-02	P0	Accept text and multiple photos with role labels	Reference and room photos are not silently treated as the same scene
FR-03	P0	Extract candidate objects and visible labels	Each observation carries source and confirmation status
FR-04	P0	Capture measurements and units	Critical dimensions require measured/user-confirmed input; unit conflicts block matching
FR-05	P0	Ask dependency-aware clarification questions	Missing critical substrate/load information prevents final fixing selection
FR-06	P0	Maintain owned tools/materials inventory	Uncertain or incompatible items are not deducted from requirements
FR-07	P0	Generate structured BOM	Every line has role, specification, quantity basis and unresolved dependencies

ID	Priority	Requirement	Acceptance criterion
FR-08	P0	Retrieve exact catalogue variants	Each recommendation links to an identifiable product and variant
FR-09	P0	Validate compatibility	Screw, fixing, bracket and bit constraints are checked against source evidence
FR-10	P0	Explain recommendation	User can see why the item fits and which facts remain unverified
FR-11	P0	Find stores by postcode or optional device location	App remains usable after location permission denial
FR-12	P0	Display stock provenance and age	Unknown, stale and out-of-stock are visibly distinct
FR-13	P0	Present basket options	Quantities, pack sizes, taxes shown by source, estimated total and exclusions are visible
FR-14	P0	Hand off to retailer	Retailer links work without implying successful reservation or purchase
FR-15	P0	Show project checklist offline	Saved checklist remains readable without network; stale data is labelled
FR-16	P0	Present safety stops and escalation	Safety-critical uncertainty cannot be dismissed by optimistic model text
FR-17	P0	Link assembly steps to manuals or reviewed templates	Applicable product and document revision remain identifiable
FR-18	P0	Provide export and project deletion	User receives a readable project export and deletion status
FR-19	P1	Show aisle/zone/rack where verified	All displayed locations match selected store and map revision
FR-20	P1	Scan barcodes during shopping	Different size/variant triggers a mismatch and revalidation
FR-21	P1	Approve a reservation integration	Exact action approved; retries do not create duplicates
FR-22	P1	Track assembly progress and deviations	Changed component invalidates affected downstream steps
FR-23	P1	Share summary with store associate	Only user-selected project information is shared
FR-24	P2	Multi-retailer basket optimisation	Hard constraints remain enforced across substitutions and split baskets
FR-25	P2	Voice interaction and room visualisation	Visualisation is labelled conceptual and cannot override measured dimensions

6. Safety requirements

Fischer's guidance links fixing selection to substrate and applied load. Bosch documents detection limits and conditions that can prevent live-wire detection. These are reasons to implement product-specific safety gates rather than trust a generated shopping list or a detector's negative indication as clearance to drill. [web:22][web:25][web:32]

ID	Required control
SR-01	Separate visual guesses from user-confirmed facts and manufacturer-supported specifications
SR-02	Do not certify wall type, allowable loads, concealed services or safe drilling locations from photos
SR-03	Require substrate, fixing system, bracket geometry, intended loading and applicable installation conditions before final mounting recommendation
SR-04	Never calculate shelf capacity by simply adding nominal anchor ratings; use applicable system evidence or refer to a qualified person
SR-05	Match detector capabilities to relevant hazards; do not describe every detector as detecting all water pipes, particularly plastic pipes
SR-06	Present limitations of detection and inconclusive results; uncertainty triggers a stop, not permission to proceed
SR-07	Preserve manufacturer warnings, installation conditions and approved combinations; never let an LLM rewrite them into weaker instructions
SR-08	Re-run safety checks after every material, dimension, load or product change
SR-09	Exclude hazardous work types and provide an appropriate professional handoff
SR-10	Have qualified reviewers approve templates and release-blocking test cases before public mounting guidance

Disclaimers do not substitute for safe design. A photographed finished shelf does not reveal internal joinery, material grade or proven capacity. The initial product should help collect evidence and identify gaps, not present itself as an engineer or inspector.

7. Example shelf project — logic, not an installation prescription

User uploads a reference shelf, installation area, a board photo and owned drill information.

The app asks: freestanding or wall-mounted? What are the measured board dimensions? What is the intended total load and load distribution? What is the confirmed substrate? Are brackets present, and what do their manuals specify? Which drill, bits and battery system are owned? What are budget, postcode and transport constraints?

The initial output can include board, support system, mounting fasteners, wood fasteners, bits, measuring/levelling tools, suitable detector and relevant PPE as categories. It must not invent exact screw/anchor sizes before compatibility evidence is available.

Inventory reconciliation example: a confirmed suitable drill is marked “owned”; an unidentified screw bag is marked “needs identification”; an unsuitable bit remains in inventory but is not counted as satisfying the project. If the wall type remains unknown, product matching for wall fixings stays blocked while the app offers a store-consultation summary or a freestanding alternative.

8. Agentic architecture

Recommended architecture

Use one stateful orchestrator with bounded specialist workers and deterministic validation. “Multiple agents” is an implementation mechanism, not the customer value proposition. Avoid free-running agent debates and unrestricted browsing or purchases.

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Mobile / responsive web interface
    -> Auth, consent, uploads and project API
    -> Durable workflow orchestrator + typed project state
        -> Visual extraction worker
        -> Requirements and inventory worker
        -> Project/BOM planner
        -> Catalogue retrieval worker
        -> Store/fulfilment worker
        -> Assembly guidance worker
        -> Deterministic compatibility + safety policy services
    -> Approved retailer/location/manual adapters
    -> Relational data + object storage + search index
    -> Evidence registry, traces, evaluation and admin review
```

Worker responsibilities

Component	Reads	Produces	Prohibited behaviour
Orchestrator	Confirmed project state	Next task, bounded retries, checkpoints	Ignoring a blocked safety state
Vision worker	Images and visible labels	Candidate objects, OCR, uncertainty	Claiming concealed facts or precise unmeasured scale
Requirements worker	User goals and observations	Clarification questions and constraints	Treating unanswered fields as confirmed
Planner	Confirmed spec and approved templates	Generic BOM and dependency graph	Inventing structural ratings
Catalogue worker	BOM specs and approved feeds	Product candidates with exact IDs	Fabricating SKUs, prices or stock
Compatibility service	Product attributes and rules	Pass / fail / needs evidence	Overriding missing safety-critical attributes
Store worker	User area and feasible products	Stores, basket options, provenance	Equating online stock with branch stock

Component	Reads	Produces	Prohibited behaviour
Navigation service	Store-specific map and locations	Zone/aisle list or route	Reusing another branch's plan
Assembly worker	Purchased variants and manuals	Sequenced steps and checkpoints	Improvising around missing warnings
Safety policy service	All state changes	Block / allow bounded next action / escalate	Treating a disclaimer as mitigation

State machine

Draft -> Needs clarification -> Specification confirmed -> BOM reviewed -> Products validated -> Shopping plan approved -> Acquiring -> Items verified -> Ready for assembly -> In progress -> Completed.

Additional states: Safety blocked, Data unavailable, Professional review required, Cancelled. Every material change invalidates affected decisions. A failed service call cannot advance the project to “verified.”

Suggested prototype stack — illustrative, not mandatory

- Responsive web app/PWA first; native application later if camera, offline and store workflows justify it.
- Typed API backend, relational database, object storage, job queue and durable workflow runtime.
- Multimodal model for extraction and conversation; retrieval over curated manuals and product records.
- Deterministic unit conversion, quantity calculations, compatibility rules and basket constraints.
- Schema-validated tool outputs, observability, feature flags, admin review and labelled evaluation dataset.
- Use vector search for candidate retrieval, not as proof of technical compatibility.

9. Data model and evidence handling

Entity	Important fields
Project	ID, owner, language, category, budget, constraints, state, safety status
Observation	Source image/text, candidate value, confidence, confirmation status
Measurement	Value, unit, method, timestamp, source, confirmed_by
OwnedItem	Model, barcode, quantity, dimensions, condition, suitability evidence
Requirement	Role, material, dimensions, compatibility constraints, dependencies
BOMLine	Requirement ID, required quantity, owned suitable quantity, purchase quantity
ProductVariant	Retailer, SKU, GTIN if available, exact attributes, pack size, manual

Entity	Important fields
Offer	Variant, seller, price, tax presentation, fulfilment, timestamp
StockObservation	Store, variant, status, source, observed_at, expiry policy
StoreLocation	Store ID, address, opening hours, coordinates, source
ProductPlacement	Store, variant, zone/aisle/rack/bin, map revision, verified_at
Basket	Offers, quantities, substitutions, totals, unresolved gaps
AssemblyStep	Dependencies, tools, warnings, manual/source, confirmation
Approval	User, action, resolved arguments, time, expiration, execution result
EvidenceRecord	Source, version, retrieval time, licence/access terms, derived claims

All derived recommendations should retain evidence lineage. Unknown is a valid field value. Stock age limits should be negotiated with each source; there is no universal freshness interval that makes inventory guaranteed.

10. Retailer integration and indoor-navigation feasibility

Existing consumer-facing retailer functions demonstrate useful data exists within their ecosystems. They do not establish that an independent app has lawful, stable or commercially acceptable access. [web:6][web:36][web:38]

Data-access levels

Level	Available basis	Product promise
A	Curated catalogue + retailer links	Planning, compatibility and retailer handoff; no live-stock promise
B	Approved catalogue/offer feed	Updated product/price matching within feed terms
C	Store-level inventory integration	Timestamped branch availability, still not a guarantee
D	Authorised product-placement data and store plan	Verified aisle/zone guidance and route planning
E	Validated indoor positioning infrastructure	Dynamic turn-by-turn positioning within supported stores

Navigation implementation

1. First show store name, department and aisle only where verified.
2. Add an indoor path graph with entrances, walkable edges, service desks, checkout and accessible routes.
3. Link each product placement to the correct store and current map version.
4. Order shopping stops heuristically and provide a text alternative to the map.
5. Use user-selected entrance or QR/barcode landmark localisation before investing in continuous positioning.

6. Handle blocked aisles, relocated products and stock discrepancies through staff fallback and reporting.
7. Offer “show this list to an associate” when placement information is unavailable.

Do not infer exact product locations from a generic floor-plan image. A department plan, a SKU-to-rack placement table and the user's current position are different datasets. Ordinary outdoor location does not establish an accurate indoor aisle position.

Retailer discovery checklist

Ask each potential partner about catalogue schemas, model/GTIN quality, stock granularity, branch IDs, map formats, placement update cadence, permitted caching, rate limits, fees, affiliate terms, deep-link support, test environments, liability allocation and data-usage restrictions. Do not base the business on undocumented internal APIs or unattended scraping.

11. Non-functional and operational requirements

Proposed pilot targets—not measured performance claims:

- Interactive acknowledgements under 3 seconds at p95; complete bounded plan under 60 seconds at p95 when dependencies respond. Show progress and partial-state recovery.
- 99.5% monthly pilot service availability, excluding explicitly identified external dependency outages from internal diagnosis, not from the user's experience report.
- 100% of recommended exact variants have a source and retrieval timestamp.
- No invented stock or aisle data in the release test suite.
- No unapproved writes or duplicate reservation execution in action tests.
- All critical safety regression cases block unsafe progression. Passing a finite test suite is not proof of real-world safety.
- German/English and metric units tested for dimensional consistency.
- Accessible contrast, scalable text, screen-reader labels and a non-map shopping route.
- Encryption, least-privilege access, upload scanning, tenant isolation, rate limiting and redacted logs.
- Treat retailer pages, uploaded images, OCR text and manuals as untrusted data, not tool instructions.
- Restrict tool permissions and outbound domains; cap model calls, retries and cost per project.
- Track unsupported requests, escalation rates, incident reports and data-source failures.

12. Privacy, legal and governance workstream

The GDPR principles include purpose limitation, minimisation, retention limitation, security and accountability. European Commission guidance also identifies privacy by design/default. EU AI Act guidance describes disclosure requirements for systems that interact with people,

such as chatbots. Detailed applicability must be reviewed against the final deployment and contractual roles. [web:66][web:67][web:68][web:21]

Proposed controls:

- Use postcode entry as an alternative to precise location; request precise location only when necessary.
- Let users crop photos and remove unnecessary faces, addresses and documents; strip unneeded metadata.
- Document lawful bases, controller/processor roles, retention, international transfers, vendor terms and deletion handling.
- Keep model training on customer home photos disabled by default; any optional use needs a separately assessed, transparent basis.
- Set short, documented raw-image retention and offer user-controlled project deletion.
- Clearly identify AI-generated guidance and distinguish verified specifications from visualisation.
- Review whether a data-protection impact assessment is needed; do not assume it is always required or never required.
- Review product advice liability, insurance, consumer law, returns, affiliate disclosure, content licences and retailer trademarks.
- Never imply retailer partnership or engineering certification without an agreement or valid basis.

13. German market analysis

Market size and definitions

BHB reports the German DIY core market at about EUR 49.10 billion in 2025. It reports total gross turnover for German DIY stores of EUR 24.67 billion, including EUR 20.58 billion in the BHB/GfK Total Store Report. These are overlapping scopes, not additive markets. [web:12]

The BHB/GfK store segment declined 1.6% in 2025. This suggests a cautious commercial environment; it does not by itself establish willingness to pay for AI assistance. [web:9][web:16]

Interpretation

Hardware spending is the relevant transaction context, not the revenue TAM of this software. A credible serviceable market requires project frequency, target-user reach, supported product categories, retailer coverage, adoption and a monetisation mechanism. No defensible software TAM or market share is asserted here.

Demand hypotheses

- Customers value fewer failed trips and fewer incompatible purchases.
- Retailers value successful project baskets and associate productivity.
- Multilingual support may improve the experience for non-native German speakers.
- Tool reuse and rental may appeal to occasional DIY users.
  All require direct testing; market size does not validate them.

Commercial options

Model	Advantage	Main constraint	Validation
Retailer-funded B2B2C pilot	Better access to authoritative product/store data	Long sales cycle and retailer lock-in	Seek one paid pilot with data access
Consumer per-project fee	Suits episodic DIY demand	Acquisition cost and willingness to pay	Test purchase intent with real completion support
Consumer subscription	Recurring revenue possibility	Users may not undertake projects frequently	Compare actual repeat usage with per-project model
Disclosed affiliate revenue	Low checkout complexity	Coverage, attribution and neutrality	Verify terms and revenue on eligible transactions
Associate-assistance licence	Fits in-store workflow	Training, integration and measurable ROI	Compare assisted sessions with baseline

Recommended exploration: one retailer-backed or independent-store pilot while preserving a retailer-neutral domain model. This is a recommendation, not evidence that retailers will participate.

Illustrative unit economics, not a forecast

Consumer scenario: 1,000 paid projects/month at EUR 4.90 yields EUR 4,900 gross receipts before tax treatment, refunds and costs. This says nothing about profit or acquisition feasibility.
Affiliate scenario: 10,000 completed plans x 20% attributed eligible orders x EUR 100 eligible basket x 3% assumed commission yields EUR 6,000/month before operating costs. Every input is hypothetical; retailer terms may make the model infeasible.
Retailer value should be measured through incremental completed project baskets, avoided returns, associate time and customer satisfaction—not attributed sales alone.

14. Competitor research

Public-source review, not a hands-on benchmark. Do not infer absent features from incomplete marketing pages.

Competitor	Concise evidence-backed profile	Relevance / limitation / response
OBI / heyOBI	Official page advertises expert digital advice, shopping lists, availability and Markt-Navi product locations. [web:6]	Strong German incumbent with native store context. Win on reviewed compatibility and continuity only if access and experience are better; navigation alone is not differentiation.
HORNBACH app	Official page describes barcode details, aisle numbers, availability, product comparison and shopping lists sorted by aisle. [web:36]	Strong project-shopping competitor. An independent app must add reliable project decomposition and owned-item reconciliation.
BAUHAUS app	Official page advertises product finder and lists for locating tools/products; official store listing describes product locations and location-sorted lists. [web:38] [web:37]	Direct competitor to the proposed in-store finding feature. Test branch coverage and placement freshness rather than assuming universal precision.
toom app	Official page advertises store availability, reservation/collection and home delivery. [web:83]	Relevant German shopping ecosystem. Photo-derived BOM and SKU-level indoor maps were not established in reviewed evidence.
hagebau app	Official page describes market search, product scanner, shopping lists, offers and customer-card features. [web:46]	Competes in trip preparation and shopping support. Verify coverage and data arrangements store by store.
IKEA app / Kreativ	Official German page describes lists, availability, easier product finding, Shop & Go and scanning/virtually redesigning a room with Kreativ. [web:88]	Strong inspiration-to-own-catalogue journey. Do not confuse room visualisation with validated mounting engineering or cross-retailer hardware selection.
Home Depot / Magic Apron	August 2026 announcement describes image uploads, text/voice, multilingual conversations, product/project help, local availability, maps and aisle/bay locations across all 2,000+ US stores. [web:81]	Closest verified end-to-end functional benchmark. US footprint limits direct German store overlap, but the capability threat is high. Full physical-safety validation and owned-inventory reconciliation are not established by this announcement.
Lowe's / Mylow	Official consumer page describes project guidance, step-by-step help and product recommendations. September 2026 partner account describes customer and associate assistants answering millions of questions since spring 2025. [web:52][web:53]	Strong conversational project and associate-assistance benchmark. German retailer-neutral coverage is not established.
DIWAI	Vendor site advertises photo-to-plan, tools/materials/costs, safety guidance, nearby UK stores, live-price shopping lists, voice and paid design mode. Listed annual prices are GBP 39.99 Pro and GBP 79.99 Premium. [web:64]	Very close concept-level competitor. Actual live-price coverage, compatibility quality and commercial adoption are unverified; validate by hands-on testing.
SnapToBuild	Vendor site advertises photo-to-materials/tools/steps, editable shopping lists, German/English, iOS 18+, three free analyses/month and a stated 90-second average analysis time. [web:65]	Especially relevant to the shelf example and German-language audience. Site mixes availability claims with “coming soon” labels, so release status needs checking. Store inventory/maps and technical verification are not established.
Craft DIY	App listing advertises photo-based woodworking analysis, material/joinery identification, build guides, cut lists and materials. [web:61]	Relevant specialist discovery competitor. Treat visual identification as advertised capability, not proven measurement or safety accuracy.

Competitive conclusion

Do not claim “first AI DIY assistant” or “no competitors.” Photo input, material lists, product advice and store wayfinding already exist in different combinations. The proposed positioning is a hypothesis: an evidence-backed, retailer-portable project record that reconciles owned components and validates shopping substitutions before assembly.

Hands-on benchmark protocol

Test ten identical projects across relevant products, with repeated runs and documented dates/versions. Include an ambiguous wall, unknown screws, incompatible bracket holes, a changed load, unavailable stock, a wrong barcode variant and a request for a pipe-safe drilling location.

Score separately: question quality; BOM completeness; unsupported specifications; compatibility; safety stops; exact product match; stock provenance; location correctness; purchased-item verification; language quality; latency; total cost. Use an independent DIY/domain reviewer. Keep vendor claims distinct from observed results.

15. Differentiation and defensibility

Candidate advantages to validate:

1. Structured compatibility graph with provenance, not just plausible text.
2. Accurate reconciliation of what the user already owns.
3. Human-confirmed facts and clear uncertainty states.
4. Store-specific evidence and honest degradation when data is unavailable.
5. Revalidation after in-store substitution.
6. Project continuity from planning through assembly and leftover inventory.
7. German/English terminology mapped to exact specifications.

Potential defensibility comes from licensed integrations, reviewed rules, annotated failure cases, trusted workflow and measurable project outcomes. An agent framework or chatbot UI alone is readily imitated. Retailer neutrality creates breadth but may weaken the incentives of a single retailer to share data; validate that tension early.

16. Roadmap with decision gates

Durations are planning estimates for a small experienced team; partner timelines can dominate.

Phase	Timing	Deliverables	Exit gate
0: Problem and access validation	Weeks 1–3	15–20 user interviews; 3–5 retailer/associate interviews; five observed shopping journeys; data-access requests; hazard inventory	Repeated user problem plus a credible legal data path for the intended pilot

Phase	Timing	Deliverables	Exit gate
1: Concierge prototype	Weeks 4–6	Clickable app, manually reviewed shelf plans, curated products, benchmark cases	Users understand uncertainty and can act on the list; no unresolved unsafe flow
2: Bounded MVP	Weeks 7–12	Project state, photo intake, clarification, inventory, BOM, evidence-backed matching, store discovery, export	Reviewed end-to-end tasks pass technical and safety tests
3: Controlled pilot	Weeks 13–18	One region/store partner, approximately 50–100 consented projects, associate fallback, incident process	Useful completion outcomes, acceptable cost and reliable supported-store data
4: Operational hardening	Months 5–6	Barcode verification, approved aisle/zone data, caching, monitoring, privacy review	Stable source updates and demonstrated advantage over ordinary retailer shopping
5: Expansion	Months 7–12	Additional categories/retailers, richer shopping optimisation, optional native app/voice	Replicable onboarding and positive or credible unit-economics path

Do not wait until phase 4 to discover that map or inventory access is impossible. Request it in phase 0 and test a sample integration before promising navigation. If no partner is available, keep Level A planning/handoff and describe navigation as a future dependency.

First six development sprints

- Sprint 1: research synthesis, hazard analysis, schemas, evidence model and clickable user flow.
- Sprint 2: intake, photo-role labels, measurement capture and clarification logic.
- Sprint 3: owned inventory, approved project templates and generic BOM engine.
- Sprint 4: curated catalogue ingestion, variant resolution and deterministic compatibility tests.
- Sprint 5: postcode/store discovery, timestamped availability/handoff and offline list.
- Sprint 6: manual-grounded assembly flow, adversarial testing, privacy controls and controlled pilot readiness.

Do not add agents merely to fill these sprints; use ordinary functions wherever deterministic code is simpler and more reliable.

17. Team and budgeting assumptions

Suggested responsibilities: product/UX lead, full-stack engineer, AI/data engineer, fractional DIY/safety specialist, and fractional privacy/security/legal support. Retailer integration ownership must be explicit.

Illustrative 12-week build: three full-time-equivalent contributors at EUR 7,000–12,000 fully loaded per person-month for three months implies EUR 63,000–108,000 labour. Add EUR 10,000–25,000 for fractional specialist work, research, devices, hosting and model usage; a 20% contingency gives roughly EUR 88,000–160,000. These are internal planning inputs, not

researched supplier quotations. Avoid double-counting specialist time already included in the full-time-equivalent assumption.

A student/founder demonstrator can have much lower cash outlay if labour is unpaid and data is curated, but that does not establish production readiness. Data licensing, indemnities, ongoing expert review and partner procurement are excluded from the illustrative budget and need separate quotes.

18. Evaluation and release readiness

Primary outcome

Percentage of eligible projects completed with the correct compatible items, no avoidable extra shopping trip and no reported safety incident. Report the three components separately as well; absence of reported incidents is not proof of safety.

Proposed pilot metrics

- At least 90% expert-reviewed BOM completeness on supported test projects.
- At least 95% exact-variant retrieval precision on the curated catalogue benchmark.
- 100% provenance for prices, availability and product placements shown.
- Zero severe unsafe recommendations in the release-blocking safety suite.
- Track time to usable list, clarification burden, plan abandonment, substitutions, repeat trips and returns.
- Track model/tool cost per completed project and expert-review time.
- Track stock/location accuracy only in stores for which a verifiable source exists.
- Compare with a baseline: user shopping independently using existing retailer tools.

These thresholds are proposed acceptance goals. Confidence intervals and sample sizes must accompany reported results, especially for rare safety events.

Critical test cases

1. Wall substrate unknown: no final anchor recommendation.
2. Photo without a scale: no exact dimensional assertion.
3. Screw appears visually similar but wrong variant: mismatch detected.
4. Intended load rises: downstream fixing decisions invalidated.
5. Store feed is stale: stale label and verification option shown.
6. Map missing: no fabricated aisle.
7. Detector cannot reliably cover relevant pipe material: no false clearance.
8. Manual contradicts generated text: manual requirement wins.
9. Malicious instructions embedded in a product page: ignored as instructions.

10. Checkout/reservation tool retries: no duplicate external action.
11. Location denied: postcode search remains available.
12. Network lost: saved list readable, new stock claims unavailable.
13. Ambiguous pack quantity: basket total withheld until resolved.
14. Product replaced during shopping: assembly plan rechecked.

19. Major risks and mitigations

Risk	Impact	Mitigation / decision
Unsafe fixing or drilling advice	Injury/property damage	Bounded scope, evidence gates, qualified review and escalation
No authorised catalogue/map access	Core shopping/navigation promise infeasible	Phase-0 partner validation; honest lower-data product mode
Incorrect exact product variant	Incompatible purchase	Typed attributes, barcode/model verification, reviewed matching
Stale stock/placement	Wasted trip and lost trust	Timestamps, expiry, store confirmation and reporting
Incumbent feature convergence	Weak differentiation	Benchmark before building; focus on validated unmet workflow
Hallucinations and prompt injection	False claims or unauthorised actions	Evidence-backed outputs, untrusted-input boundaries, least privilege
Privacy leakage from home photos	Personal harm and compliance exposure	Minimise capture, restrict access, retention/deletion controls
Low repeat usage	Poor subscription economics	Test project fees and B2B2C alternatives
Affiliate bias	Loss of trust	Separate feasibility filtering from commercial ranking; disclose incentives
Integration overhead	Slow rollout	Common domain model and capability-based adapters

20. Immediate next actions

1. Confirm whether this is an academic prototype, founder MVP or retailer proposal.
2. Pick one city/region and one or two tightly bounded shelf/furniture workflows.
3. Interview target users using their real last shopping project, not only hypothetical interest.
4. Request sample catalogue, stock and placement data from potential partners.
5. Build a 30–50-case expert-reviewed test set, including failure and uncertainty cases.
6. Run the same shelf cases through competitor tools where access permits.
7. Implement a manual-first concierge version before automating exact product advice.
8. Decide whether to proceed based on demonstrated user value, safe behaviour and feasible data access.

21. Questions that materially change the next version

- Is the intended deliverable an MBA/academic prototype, an investable startup MVP or a proposal to a retailer?
- Which initial city/country and primary user language should the pilot serve?
- Do you already have contacts or data access at OBI, toom, hagebau, BAUHAUS, IKEA or an independent shop?
- What team, budget and target demonstration date are available?
- Is retailer checkout handoff sufficient initially, or is reservation/purchase integration mandatory?
- Would a bounded furniture/freestanding-shelf first release be acceptable before general wall mounting?